

Gene Regulation and Systems Biology

NetPerturb: a reproducible Nextflow pipeline for in silico perturbation of single-cell gene regulatory networks from a Seurat object

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Abstract

Motivation: Predicting how a genetic or pharmacological perturbation affects each cell population in a single-cell RNA-seq (scRNA-seq) dataset is a common question, but in silico perturbation tools based on gene regulatory networks (GRNs) are hard to install, scale poorly to large atlases, and return results that are difficult to compare across methods and to share.

Results: We present NetPerturb, a Nextflow pipeline that starts from a processed Seurat object and a list of target genes and returns a single, self-contained HTML report. NetPerturb balances cell identities by downsampling, checks the targets before any network is built, and infers one GRN per identity with GENIE3, scRank or hdWGCNA. It then ranks identities by a target-perturbation score. In parallel, it can run a virtual knockout with scTenifoldKnk, including joint knockouts of several genes, followed by gene set enrichment of the genes the knockout affects. Every stochastic step is seeded from one parameter and every tool runs in a versioned container, so runs are reproducible across HPC and cloud environments. On unstimulated PBMCs, interferon-stimulated genes scored highest in monocytes, a signal that remains after correcting for a cell-type-wide offset, and building the wild-type network once per identity cut the knockout cost about 24-fold.

Availability and implementation: NetPerturb is freely available under the [TODO: MIT] licence at <https://github.com/MCBLab/NetPerturb>, with documentation, test profiles and container recipes. [TODO: Zenodo DOI of the release.]

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Supplementary information: Supplementary data are available at Bioinformatics Advances online.

Keywords single-cell RNA-seq, gene regulatory network, in silico perturbation, virtual knockout, Nextflow

Introduction

Single-cell RNA-seq (scRNA-seq) resolves tissues into cell identities (cell types, clones, or treatment states), and the next question is usually which of those identities depend on a given gene or drug target. Pooled CRISPR screens with single-cell readout answer this experimentally (Dixit et al., 2016), but they are costly and cannot be applied to archived or clinical samples. Computational alternatives therefore perturb a gene regulatory network (GRN) inferred from unperturbed cells and read the effect off the network: CellOracle simulates shifts in cell identity (Kamimoto et al., 2023), scRank ranks cell types by how much a drug target reshapes their network (Li et al., 2024), and scTenifoldKnk performs a virtual knockout and reports the genes whose regulatory context changes (Osorio et al., 2022).

In practice, using these methods on a new dataset takes considerable engineering work. Each method expects its own input format and gene universe, and memory grows quickly with the

number of cells and genes. Targets that are not expressed in some identities can end a run late in the analysis. Results from different network inference methods are rarely placed side by side, and many steps are stochastic, so the numbers are hard to reproduce. [TODO: Add 1–2 sentences on the specific gap your lab hit.]

NetPerturb addresses this with a single Nextflow (Di Tommaso et al., 2017) workflow, organised following nf-core practice (Ewels et al., 2020). Its input is the object most analysts already have, a processed Seurat object (Hao et al., 2023). It parallelises network inference and scoring across identities and targets, and it gathers every result, together with the quality control and provenance needed to interpret it, into one portable report.

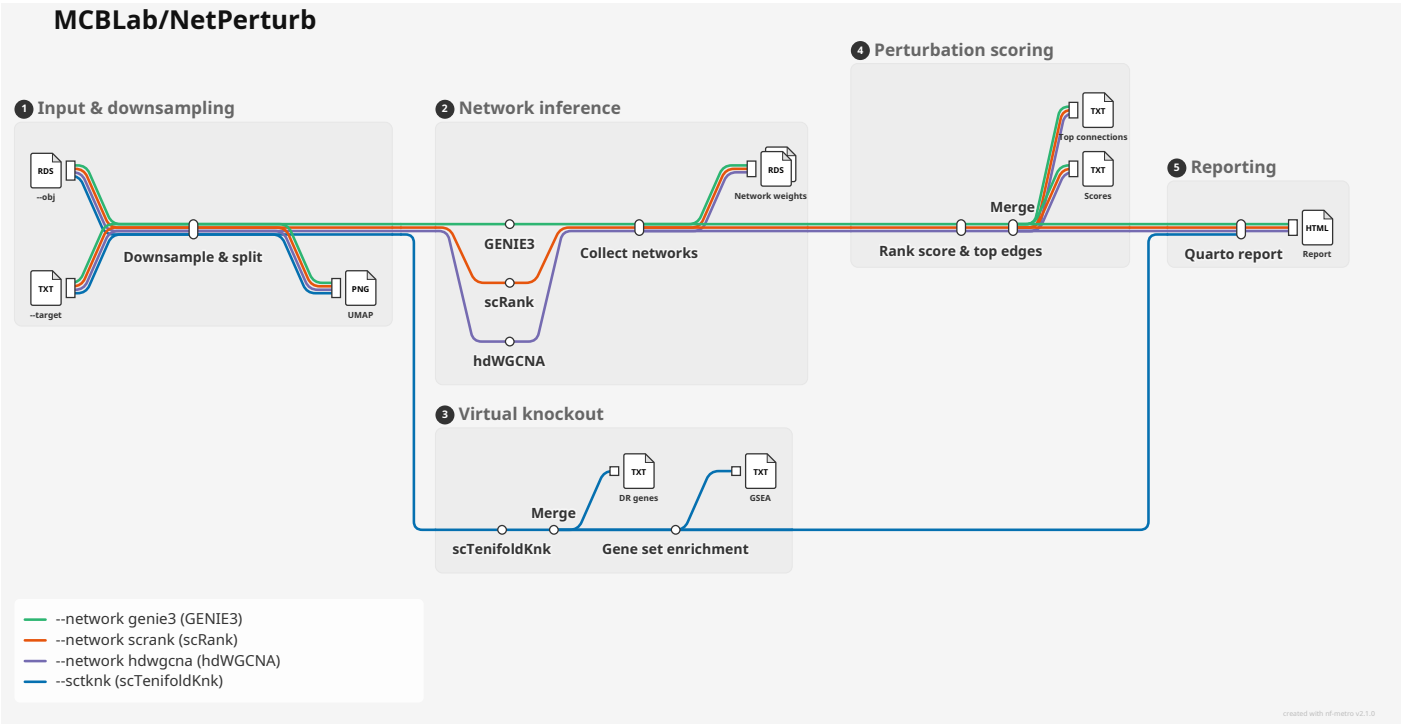


Figure 1 Overview of NetPerturb. A processed Seurat object and a list of target genes enter DOWNSAMPLE, which balances cell identities and checks the targets. The rank-score track (warm lines; one of GENIE3, scRank or hdWGCNA, chosen with `--network`) infers one GRN per identity and scores every identity $\times$ target pair. The knockout track (blue; `--sctknk`) builds one wild-type scTenifoldKnk network per identity and runs one virtual knockout per identity $\times$ target, optionally followed by fgsea. Both tracks converge on a single self-contained HTML report.

Implementation

Input and pre-processing

NetPerturb requires four inputs: a Seurat object (`--obj`), the metadata column defining cell identities (`--column`), the species, and a text file of targets (`--target`), one per line. Genes joined by a semicolon (e.g. CDK8;CDK19) are treated as a single combined perturbation. The DOWNSAMPLE step randomly keeps at most `--n_cells` cells per identity, so that every identity contributes a comparable number of cells, and drops identities that end up with fewer than `--min_cells` cells. It also builds a shared gene universe from the highly variable genes, transcription factors, drug targets and the requested targets. Each target is checked against the retained cells before any network is built. Genes that are absent from the object or have no counts are removed and recorded, a combined target keeps whichever of its genes pass, and the run stops only when no target passes. Problems that would otherwise end a run hours later are therefore caught at the start and listed in the report.

Rank-score track

For each identity, NetPerturb infers a GRN with one of three interchangeable methods (Table 1). GENIE3 (Huynh-Thu et al., 2010) and scRank’s own network builder (Li et al., 2024) are used as published. hdWGCNA (Morabito et al., 2023), which builds on WGCNA (Langfelder and Horvath, 2008), produces an unsigned co-expression network on metacells, so NetPerturb adapts its topological overlap matrix to the input scRank expects in four steps. It restores

Table 1 Network inference methods available in NetPerturb.

Method	Network	Output	Track
GENIE3	Tree-based regression	Directed weights	Rank score
scRank	scRank builder	Directed weights	Rank score
hdWGCNA	Metacell TOM + sign	Signed, sparsified	Rank score
scTenifoldKnk	PCR + tensor denoising	WT vs. KO manifold	Knockout

TOM, topological overlap matrix; PCR, principal component regression; WT, wild type; KO, knockout.

the sign of each edge from the correlation between the two genes, pads every identity to the same gene set, removes edges below the `--cut_ratio` quantile of absolute weight, and rescales the weights to $[-1, 1]$. Each identity $\times$ target pair is then scored with scRank, which compares each network before and after removing the target’s edges and returns a perturbation score in either antagonist or agonist mode. The strongest `--top_connections` edges of each target are also exported. Networks and scores are computed as separate parallel tasks and merged into one table afterwards.

Virtual knockout track

When `--sctknk` is set, a scTenifoldKnk track (Osorio et al., 2022) runs alongside the rank-score track, or on its own. The expensive wild-type network does not depend on the target, so NetPerturb builds it once per identity (SCTENIFOLDKNK_BUILD). Each identity $\times$ target knockout then runs as a separate task on that stored network (SCTENIFOLDKNK_KO). With a single build per identity, one wild-type network can serve any number of knockouts, and the knockout of a

combined target can zero the edges of all its genes at once, which the original single-gene interface does not allow. Differential regulation uses the statistic of `scTenifoldKnk` v1.0.3, which leaves the knocked-out genes out of the reference mean. [TODO: Briefly justify, e.g. “In our tests, including them inflated the mean so much that almost no gene reached significance (1 vs. 12 genes at FDR < 0.05 on a 5,600-gene network).”] If a GMT file is supplied, genes are ranked by $\log_2$ FC and tested with `fgsea` (Korotkevich et al., 2016) against, for example, MSigDB collections (Liberzon et al., 2015). Gene sets are read from local files, so the step also runs on compute nodes without internet access.

Report

The **REPORT** step renders a single self-contained Quarto HTML file ([TODO: Supplementary Fig. S1: report screenshots]). The report contains: (i) an overview recording the network method, the targets removed by quality control, the number of cells and genes available to each identity’s network, and a UMAP of the retained cells; (ii) a heatmap of target expression per identity, so that a low score can be checked against a target that is barely expressed; (iii) a searchable table and a heatmap of perturbation scores; (iv) interactive knockout networks and a table of differentially regulated genes; (v) an enrichment dot plot and table; and (vi) provenance, with the command line, software versions, parameters and the running time of each step taken from the Nextflow trace.

Reproducibility and deployment

Every tool runs in a versioned Singularity/Docker container (Kurtzer et al., 2017), so no R dependencies need to be installed. A single `--seed` parameter sets the random seed of every step that uses one: downsampling, GENIE3 (via `doRNG`), `hdWGCNA` metacells, `scTenifoldKnk` and `fgsea`. `scRank`’s network builder is the one exception, because it sets its own seed internally. Memory and CPU requests are defined per process and can be adjusted in a configuration profile. The pipeline is tested with `nf-test` at the module and workflow level, and it provides small and dataset-scale test profiles (`-profile test`).

Results

We benchmarked `NetPerturb` on the unstimulated (control) peripheral blood mononuclear cells of Kang et al. (2017), scoring 25 targets in agonist mode with `hdWGCNA` networks and running the `scTenifoldKnk` track with GO gene sets (MSigDB C5). The targets fall into three groups: the type I interferon (IFN) receptor and signalling cascade (IFNAR1, IFNAR2, JAK1, TYK2, STAT1, STAT2, IRF9 and the joint targets IFNAR1;IFNAR2, JAK1;TYK2, STAT1;STAT2); ten interferon-stimulated genes (ISGs); and five lineage genes used as controls (CD19, MS4A1, IL4R, TLR4, CSF1R). With `--n_cells` 2,000 and `--min_cells` 500, six identities were kept (7,502 cells) and two were dropped automatically (dendritic cells, 258 cells; megakaryocytes, 63 cells). Every target passed QC.

Monocytes rank first, but partly for every target.

Monocytes had the highest score for 20 of 25 targets, including all 10 ISGs (Fig. 2A). This agrees with circulating monocytes carrying the highest baseline ISG expression in these data and being the main IFN-responsive population in PBMCs. However, when log scores

are decomposed into target and identity effects, both monocyte populations score about 7-fold higher than the other identities regardless of target (+0.84 and +0.87 $\log_{10}$ relative to B cells). This offset also lifts the lineage controls: FCGR3A⁺ monocytes rank first for CD19. Only after the offset is removed does the ISG signal become specific. ISGs keep a monocyte excess of 0.39 $\log_{10}$ (Wilcoxon $P = 0.009$), whereas the upstream cascade shows none (−0.14) and the lineage controls fall below it (−0.61) (Fig. 2B). The networks themselves support this: 67% of the 15 strongest partners of an ISG in monocytes are other ISGs, against 25% in other identities and 1% for lineage genes (Fig. 2C). Scores correlate moderately with target expression (Spearman $\rho = 0.49$), but the monocyte term remains after adjusting for expression (0.77 $\log_{10}$, $P = 6 \times 10^{-9}$). Targets with no counts in an identity score at the numerical floor ($< 10^{-10}$, 9 pairs), as expected for an isolated node. A further 12 pairs whose target is detected in fewer than 1% of cells still received ordinary scores, and one of them (CD19 in FCGR3A⁺ monocytes) ranked first. [TODO: Validate against the stimulated half of the data: correlate per-identity scores with the number of IFN- β -induced DE genes per cell type from the stimulated vs. control comparison.]

The knockout track is largely target-independent in this run.

All 141 identity $\times$ target knockouts produced at least one differentially regulated (DR) gene at FDR < 0.05 (1,858 in total). The hits, however, barely depended on the target. In four of six identities the DR set was identical for most targets (median pairwise Jaccard 1.0; Fig. 2D). In CD14⁺ monocytes all 25 knockouts returned the same two genes, FTH1 and FTL, and the full FC rankings were highly correlated across targets (median Spearman $\rho = 0.74$ –0.96). Ribosomal protein genes made up 37% of hits, but only 3.2% of tested genes. The GO enrichment did not recover interferon signalling: the IFN-related sets that passed FDR < 0.25 had negative NES, mostly in B cells. We read this as the knockout signal being dominated by genes whose network position differs between the two aligned networks for reasons unrelated to the target. [TODO: Diagnose before submission: compare with `scTenifoldKnk()`’s own statistic (v1.1) on the same networks, a random-gene knockout null, and a larger `--sctknk_min_pct`; decide whether the v1.0.3 reference mean is responsible.]

Cost.

The whole run took 5.0 h of compute across all tasks (Fig. 2E). Network inference (`hdWGCNA`, about 1 min per identity) and scoring (about 31 s per target) were small next to the `scTenifoldKnk` wild-type build (43 min per identity). Building that network once per identity and reusing it for each of the 150 knockouts (5.5 s each) took 4.5 h, whereas one `scTenifoldKnk()` call per identity $\times$ target pair would have taken an estimated 108 h, about 24-fold more.

Conclusion

`NetPerturb` takes an analyst from a Seurat object to a shareable report on GRN-based in silico perturbation with a single command. It runs several network inference methods and a virtual knockout under one interface, and its seeding, containers and provenance make each result reproducible. The benchmark also shows how these scores should be read. Perturbation scores carry an identity-wide offset, so they should be compared against control targets from the same run

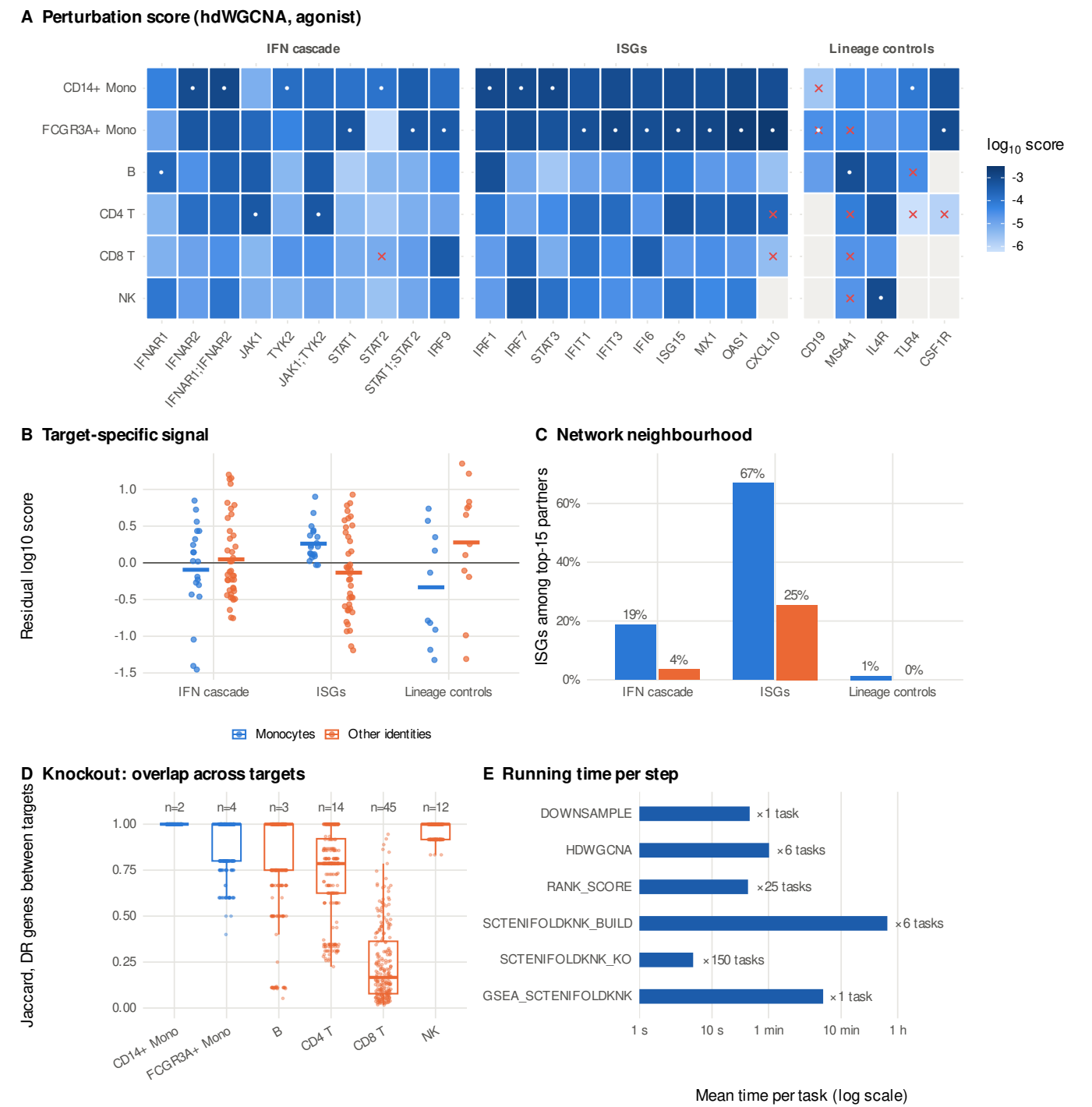


Figure 2 Benchmark on unstimulated PBMCs (Kang et al., 2017) (hdWGCNA networks, agonist mode, seed 1). (A) $\log_{10}$ perturbation score for each identity $\times$ target pair. A white dot marks the top identity for each target; a red $\times$ marks a target detected in $< 1\%$ of the identity's cells; grey marks scores at the numerical floor ($< 10^{-10}$). (B) Residual $\log_{10}$ score after removing the target and identity main effects, by target group; bars show means. (C) Share of each target's 15 strongest network partners (*top_connections*) that are ISGs. (D) Pairwise Jaccard index of the scTenifoldKnk DR gene sets (FDR < 0.05) between targets within each identity; *n*, median DR genes per target. (E) Mean running time per task for each step, from the Nextflow trace.

rather than taken at face value; scores for targets detected in very few cells are unreliable; and knockout results that do not change with the target need a null model before they are interpreted. Scores are comparable only within one inference method, and knockout effects

describe network structure, not the direction of expression change. [TODO: Future work: e.g. built-in control targets and a per-identity null, detection filter on targets, AnnData input, nf-core submission.]

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Author contributions

[TODO: CRediT roles, e.g. D.M.-C.: Conceptualization, Software, Writing – original draft. ...]

Conflict of interest

None declared.

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Data availability

The source code, container recipes and test data definitions are available at <https://github.com/MCBLab/NetPerturb> (release [TODO: vX.Y.Z], archived at Zenodo [TODO: DOI]). The benchmark uses the data of Kang et al. (2017) (GEO accession GSE96583); the processed Seurat object and the benchmark outputs are available at [TODO: Zenodo DOI].

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